Stable infinity of $\mathcal{T}$: (Question from exam - own solution)
Let $\Sigma = \{f, h\}$ be a signature, where f is a binary function symbol and h is a unary function symbol. Let $\mathcal{T}$ be the theory axiomatized by the axioms:

$$x = f(h(x), y)$$
$$y = h(x)$$

where x, y are variables and thus the above axioms are implicitly quantified. The symbol $=$ denotes the standard equality predicate, axiomatized accordingly by the axiom of the theory of equality $\mathcal{T}_E$.
Is $\mathcal{T}$ stably infinite? If so, prove it. If not, disprove it.

$\mathcal{T}$ is stable infinity, if for every formula $F \in \mathcal{T}$ there exists an $\mathcal{T}$-interpretation, where:
- $I \models F$
- I has a domain of infinite cardinality

I must also entail $\mathcal{A}$, i.e.:
- $\forall x, y : x = f(h(x), y)$ //(universal quantification is implicit)
- $\forall x, y : y = h(x)$

Given the second axiom, it must hold that $\forall x : x = h(x)$ and therefore $\forall x, y : x = h(x) \wedge y = h(x) \implies x = y$. Any interpretation can only garantee this to be the case, if the domain consists of only one element. $\mathcal{T}$ therefore is not stably infinite.

Stable infinity of $\mathcal{T}$: (Question from exercise sheet - multiple solution)
Let Σ be an arbitrary signature. Let T be the theory of equality for Σ such that for every T-satisfiable formula $F \in \mathcal{T}$ and for every finite $\mathcal{T}$-interpretation I with $I \models F$, there exists a fininite $\mathcal{T}$-interpretation J such that $J \models F$ and $|J| = |I| + 1$. Here, $|J|$ and $|I|$ respectively dnote the cardinalities of the domains of J and I.

Is T a stably infinite theory?

- **Sample Solution:** Let F be a $\mathcal{T}$-satisfiable formula and let I be a finite $\mathcal{T}$-interpretation of F. By assumption, F also has a finite $\mathcal{T}$-interpretation J such that $|J| = |I| + 1$. Morever, using the assumption and by induction on $|I|$ we have that F has a model of any finite cardinality $\mathcal{C} > |I|$. By compactness of FOL, F then has a countably infinite model and hence T is stably infinite.

- **Own solution:** The compactness theorem states, that any set of formulas Γ has a model if every subset $\Gamma' \subseteq \Gamma'$ has a model.

We can now create the set $\Gamma = \{F\} \cup \{c_i \neq c_j \,|i \neq j\}$ for infinitly many constants c_i. As F is $\mathcal{T}$-satisfiable under I and for every $I \models F$ there exists an interpretation J with $|J| = |I| + 1$ such that $J \models F$, we can inductivly prove the satisfiability of every subset $\Gamma' \subseteq \Gamma$:
- For any $\Gamma' = \{F, c_i \neq c_j\}$, it holds that $I \models F \to J \models F$. Since J has an element more to which it can map c_i, such that $c_i \neq c_j$.
- For any $\Gamma'' = \Gamma' \cup \{c_{i'} \neq c_{j'}\}$, it holds that $J \models F \to J' \models F$, as all disequalities of Γ' are already satisfied, one only needs one more element (that being the one added with $|J'| = |J| + 1$) to satisfy the newly added disequality.
  An interpretation I statisfying a set of n disequalities then needs at least n elements in its domain.

By the compactness theorem and all subsets $\Gamma' \subseteq \Gamma$ being satisfiable there exists an interpretation $I \models \Gamma$, but as Γ has infinitly many disequalities, I needs to have a domain of infinite cardinality.
Thus, $\mathcal{T}$ is stably infinite.

Non convexitivity of $\mathcal{T}_A$: (Question from exercise sheet - sample solution)
Show that the theory of $\mathcal{T}_A$ of arrays is not convex. Provide a counterexample to convexity.

Let F be a $\mathcal{T}_A$-formula $read(write(A, x, v), y) = v$. By the axioms of $\mathcal{T}_A$, we have

$$F \to (x = y \vee read(A, y) = v)$$

//either $A[y]$ was already v or $x = y$, in which case $write(A, x, v)$ this would have rewrite $A[y] = v$

However, it doesn't hold that $F \to (x = y)$, nor $F \to (read(A, y) = v)$.

Class of interpretations (1) : (Exercise sheet - sample solution)
Consider the following $\mathcal{T}_E$-formula:

$$x = y \vee y = z \vee x = z$$

where x, y, z are variables. Describe the class of all $\mathcal{T}_E$-interpreations that make this formula valid.

A $\mathcal{T}_E$-interpretation makes this formula valid if and only if the domain of the sort x contains at most two elements.

//Based on the pigeonhole principle, for any three variables x, y, z it holds that $x \neq y \wedge y \neq z \to x = z$ as x and z can only map to the one remaining element to which y has not been mapped to

Class of interpretations (2) : (Question from exercise sheet - sample solution)
Consider the following $\mathcal{T}_E$-formula:

$$x = y \to (x = a \vee x = b)$$

where x, y, z are variables and a, b are constants. Describe the class of all $\mathcal{T}_E$-interpreations that make this formula valid.

A $\mathcal{T}_E$-interpretation makes this formula valid if and only if the domain of the sort x, a, b are the same and either
1. has two elements with $I(a) \neq I(b)$
2. has one element

//For case 1, based on the pigeonhole principle, for any variables x it holds that $x \neq a \wedge a \neq b \to x = b$ as x and b can only map to the one remaining element to which a has not been mapped to

//For case 2, the validitiy trivialy holds

// Note that for any other class of interpretations the formula can not be valid as $x = a \vee x = b$ must hold, given that for any class of interpretation an interpretation I may arbitrarly map x, y e.g. $I(x) = I(y)$

Satisfiability of a $\mathcal{T}$ using Nelson-Oppen : (Question from exam - sample solution)
Let F be the formula

$$L_1 \wedge L_2 \wedge L_3 \wedge (\neg L_4 \vee \neg L_5)$$

where

L_1 denotes $f(a+2) = f(b)$ L_2 denotes $a = b - 3$ L_3 denotes $1 = read(write(A, f(a), a), f(b))$ L_4 denotes $f(a) = f(b)$ L_5 denotes $read(A, f(b-1)) = read(A, f(b))$
with a, b being integer constants, f being a unary function, A being an integer array and $+, -, 1, 2, 3$ being interpreted in the standard way over the integers. Further, read and write are interpreted over the theory of arrays.

When considering the propositional structure of F, assume that DPLL returns the following truth assingments over the literals of F :

$I_1 :\quad L_1, L_2, L_3, L_4 \mapsto \top$ and $L_5 \mapsto \bot$ $\qquad$ $I_2 :\quad L_1, L_2, L_3, L_5 \mapsto \top$ and $L_4 \mapsto \bot$

For *each* $i = \{1, 2\}$, is F satisfiable using the literal assignemnt of I_i? If it is, provide a *theory model* of F. In case F is not satisfiable, describe the *next step* for checking the satisfiability of F using DPLL(T).

When reasoning about theory literals to determine satisfiability of$F$, use the Nelson-Oppen decision procedure to reason in the combination of theories. Use the decision procedures for the theory of uninterpreted functions and the one for arrays, and use simple mathematical reasoning for deriving new equalities among the constants in the theory of linear integer arithmetic. Provide sufficient details on your steps.

First we apply renaming to make the terms theory disjoint. $L_1 := f(c_1) = f(b)$ $L_2 := a = b - 3$ $\underbrace{L_3 := c_2 = read(write(A, c_3, a), c_4)}_{\text{1 is a constent and therefore a term of } \mathcal{T}_E \text{, which is why it needs to be substituted}}$ $L_4 := f(a) = f(b)$ $L_5 := read(A, c_5) = read(A, c_4)$

For I_1 :

unintepreted functions	linear arithmetic	arrays
$f(c_1) = f(b)$	$c_1 = a + 2$	$c_2 = read(write(A, c_3, a), c_4)$
$f(a) = f(b)$	$c_2 = 1$	$\underbrace{read(A, c_5) \neq read(A, c_4)}_{\neq \text{ because of } \neg L_5}$
$c_3 = f(a)$	$c_6 = b - 1$	
$c_4 = f(b)$	$a = b - 3$	
$c_5 = f(c_6)$		

We now iteratively use the different theory solvers to derive new equalities:
- **Reasoning over Arrays** :
  $read(A, c_3) \neq read(A, c_5) \implies c_3 \neq c_5$ //note that disequalties are not shared
- **Reasoning over linear Artihmetics** :
  $c_1 = a + 2 \wedge a = b - 3 \implies c_1 = b - 1$ //added to linear arithmetic
  $c_1 = b - 1 \wedge c_6 = b - 1 \implies c_1 = c_6$ //added to shared equalities
- **Congruence Closure for $\mathcal{T}_E$** :
  $\{\{f(c_1)\}, \{f(b)\}, \{f(a)\}, \{c_3\}, \{c_4\}, \{c_5\}, \{f(c_6)\}, \underbrace{\{c_6\}}_{\text{added by shared equalities}} \}\}$

  $\{\{f(c_1), f(b), f(a), c_3, c_4, c_5, f(c_6)\}, \{c_6\}\}$

  // each equality causes a merge of the sets
  // through shared equality $c_1 = c_6$ and $\mathcal{A}_E$ we derive $f(c_1) = f(c_6)$ and therefore get the set $\{\dots, c_3, c_5, \dots\}$ which means that $c_1 = c_6$ is added to the shared equalities

  $\{\{f(c_1), f(b), f(a), c_3, c_4, c_5, f(c_6), c_6\}\}$

  //Each pair of elements in the same set represent an equality, we therefore derive $c_3 = c_6$ (as they are in the same set) and this to the shared equality
- **Reasoning over Arrays (cont'd)**
  Given the new shared equalitiy $c_3 = c_5$ we get the following contradiciton: $c_3 = c_5 \wedge c_3 \neq c_5$, thus the formula is unstatisfiable.

For I_2 :

unintepreted functions	linear arithmetic	arrays
$f(c_1) = f(b)$	$c_1 = a + 2$	$c_2 = read(write(A, c_3, a), c_4)$
$\underbrace{f(a) \neq f(b)}_{\neq \text{ because of } \neg L_4}$	$c_2 = 1$	$read(A, c_5) = read(A, c_4)$
$c_3 = f(a)$	$c_6 = b - 1$	
$c_4 = f(b)$	$a = b - 3$	
$c_5 = f(c_6)$		

We now iteratively use the different theory solvers to derive new equalities:
- **Reasoning over linear Artihmetics** :
  $c_1 = a + 2 \wedge a = b - 3 \implies c_1 = b - 1$ //added to linear arithmetic
  $c_1 = b - 1 \wedge c_6 = b - 1 \implies c_1 = c_6$ //added to shared equalities
- **Congruence Closure for $\mathcal{T}_E$** :
  $\{\{f(c_1)\}, \{f(b)\}, \{f(a)\}, \{c_3\}, \{c_4\}, \{c_5\}, \{f(c_6)\}, \underbrace{\{c_6\}}_{\text{added by shared equalities}} \}\}$

  $\{\{f(c_1), f(b), c_4, c_5, f(c_6)\}, \{f(a), c_3\}, \}$

  // each equality causes a merge of the sets
  // through shared equality $c_1 = c_6$ and $\mathcal{A}_E$ we derive $f(c_1) = f(c_6)$
  // through congruence closure we derive shared equality $c_4 = c_5$
- No more (dis)equalties can be derived and no contradiction exists, so the formula is satisfiable:
- **Constructing an interpretation $I \models (L_1 \wedge L_2 \wedge L_3 \wedge \neg L_4 \wedge L_5)$** :
  $-c_3 \neq c_4 \implies 1 = read(A, c_4) = read(A, f(b)) \implies I(A[f(b)]) = 1$
  //nothing else is fixed and can therefore be arbitrary assigned, so e.g.:
  $-I(a) = 0, I(b) = 3 \implies I(f(b)) = I(f(3)) = 0$
  $-I(f(a)) = I(f(0)) = 1$ //or any value other then 0 as $f(a) \neq f(b)$
  $-\forall x \neq 1 : I(f(x)) = 0, \forall x : I(A[x]) = 1$

Formula Validitiy : (Question from exercise sheet - sample solution)
Consider two axiomatizable theories $\mathcal{T}_1$ and T_2 such that $\Sigma_{\mathcal{T}_1} = \Sigma_{\mathcal{T}_2} = \Sigma$ and $\mathcal{A}_{\mathcal{T}_2} \subseteq \mathcal{A}_{\mathcal{T}_2}$. Recall that $\Sigma_{\mathcal{T}_i}$ and $\mathcal{A}_{\mathcal{T}_i}$ denote, respectively, a signature and set of axioms defining $\mathcal{T}_i$, for $i = 1, 2$. Let F be a formula over signature Σ.
- *a*) If F is valid in $\mathcal{T}_1$, is it also valid in $\mathcal{T}_2$?
- *b*) If F is valid in $\mathcal{T}_2$, is it also valid in $\mathcal{T}_1$?

- **a)** No, as can be seen with the following counterexample:
  Let $\mathcal{A}_{\mathcal{T}_2} = \emptyset, \mathcal{A}_{\mathcal{T}_1} = F$, then formula F is valid in $\mathcal{T}_1$ but not in $\mathcal{T}_2$ as every interpretation of $\mathcal{T}_1$ must satisfy F by the axiom $\mathcal{A}_{\mathcal{T}_1}$, where as $\mathcal{T}_2$ has no such restrictions
- **b)** Yes, as every $\mathcal{T}_1$-interpretation must also be a $\mathcal{T}_2$-interpretation for which we know, that they all statisfy F.

Finding a model : (Question from exercise sheet - sample solution)
Find a model of the following formula in the $\mathcal{T}_E \cup \mathcal{T}_A \cup \mathcal{T}_Z$

$$read(A, f(x, y)) = x + y \wedge read(A, f(y, x)) \neq y + x$$

For our model we consider the domain $\mathbb{Z}$ of integers. We then interpret arrays as functions on $\mathbb{Z} \mapsto \mathbb{Z}$ and define $read^I$ as the function:

$$read(B, z)^I = B(z^I) \text{ for all } B, z$$

We then define I to be;

$$x^I = 1,\ y^I = 0,\ \forall u^I, v^I \in \mathbb{Z} : f^I(u^I, v^I) = u^I,\ \forall u^I \in \mathbb{Z} : A^I[u^I] = A(u^I) = u^I$$

It then follows that:

$$f(x, y)^I = 1,\ f(y, x)^I = 0,\ (x + y)^I = 1,\ (y + x)^I = 1,\ read(A, f(x, y)^I) = 1$$
$$read(A, f(x, y)^I) = 0$$

And thus: $I \models read(A, f(x, y)) = x + y \wedge read(A, f(y, x)) \neq y + x$

Stable infinity of $\mathcal{T}$: (Question from exam - own solution)
Let Σ ve an arbitrary signature. Let $\mathcal{T}$ be a first-order theory for Σ such that $\mathcal{T}$ is convex and every model of $\mathcal{T}$ has a domain of at least two elements.

Let $F \in \mathcal{T}$ be a conjunction of finitetly many $\mathcal{T}$-literals. We denote F^- the conjunction of all negative equaional literals (disequalities) in F and let F^+ be the conjunction of all remaning literals in F. That is, F^+ is the conjunction of all those $\mathcal{T}$-literals of F that do not occur in F^-.

Assume that $\mathcal{T} \models F \to x = y$. Prove that F is either $\mathcal{T}$-unsatisfiable or $\mathcal{T} \models F^+ \to x = y$. Recall that variables and hence x, y are implicitly universally quantified.

- Case 1: F is $\mathcal{T}$-unsatisfiable. In this case the entailment $\mathcal{T} \models F \to x = y$ trivially holds.
- Case 2: Assuming that $\mathcal{T} \models F$ and $\mathcal{T} \models F \to x = y \iff (F^+ \wedge F^-) \to (x = y) \iff \neg F^+ \vee \neg F^- \vee (x = y) \iff F^+ \to (\neg F^- \vee (x = y))$

  As $F^- = \bigwedge_{i=1}^{t} (x'_1 \neq j'_1)$ we can rewrite $\mathcal{T} \models F \to x = y \iff \mathcal{T} \models F^+ \to ((x'_1 = y'_1) \vee (x'_2 = y'_2) \vee \ldots \vee (x = y))$. Using the convexity of $\mathcal{T}$ we know that $F \to e$ for an equality $e \in \{(x = y), (x'_1 = y'_1), \dots\}$ which we can rewrite as $\neg F \vee e$. We now have to consider two cases:
  Case 1: $\mathcal{T} \models F^+ \to x = y$ in which case we are done

  Case 2: $\mathcal{T} \models F^+ \to (x'_i = y'_i) \iff \mathcal{T} \models \neg F^+ \vee (x = y)$ for any $(x'_i \neq y'_i) \in F^-$. But since $\mathcal{T} \models F$ it follows that $\mathcal{T} \models F^+$, therefore the formula can only be satisfiable if $\mathcal{T} \models x'_i = y'_i$ but as $\mathcal{T} \models F^-$ and $(x'_i \neq y'_i)$ is a clause of F^- it follows that $\mathcal{T} \not\models x'_i = y'_i$. As this case results in a contradiction only case 1 can be there and therefore it holds that $(\mathcal{T} \models F \to (x = y) \wedge \mathcal{T} \models F) \implies \mathcal{T} \models F^+ \to x = y$